LECTURE 4

SELF-SUPERVISED LEARNING AND FOUNDATION MODELS

Geospatial Representation Learning

UNIVERSITÄT BONN

Learning Outcomes

Lecture

- Explain the principles of self-supervised learning for geospatial and Earth observation data.
- Compare contrastive, masked, generative, and multimodal pretraining strategies.
- Describe how geospatial foundation models are trained and adapted to downstream tasks.
- Critically assess the benefits and limitations of foundation models for geospatial applications.

Lab

- Use pretrained geospatial or vision foundation models to extract representations from geospatial data.
- Apply extracted embeddings to a downstream task such as classification, clustering, or retrieval.
- Compare pretrained representations with task-specific baseline features.
- Discuss transferability, domain shift, and evaluation challenges of pretrained geospatial models.

PRACTICAL 4

USING PRETRAINED GEOSPATIAL MODELS

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